

Electromagnetism (BECS 21422)

Course Introduction

(Vector analysis, electrostatics, magnetostatics and electromagnetism)

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Course Organization and Class Policies

□ Evaluation

- Final exam (Answer any 4 of 5 questions) : 90%
- Class participation and tutorials: 10%

□ Recommended Textbooks:

- Sadiku, M. N. O., (2014), Elements of Electromagnetics, 6th Edition, Oxford University Press
- Purcell, E. M. & Morin, D. J., (2013), Electricity and Magnetism, 3rd Edition, Cambridge University Press
- Hayt, W. & Buck, J., (2019), Engineering Electromagnetics, McGraw-Hill
- Griffiths, D J., (1998), Introduction to Electrodynamics, 3rd Edition, Prentice Hall of India



NOTE - Ensure to take your own notes during the lecture.

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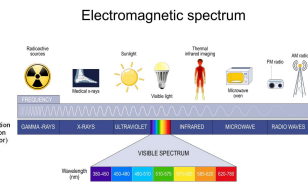
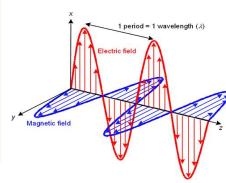
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Course Aim/ Objectives

At the completion of this course students will be able to:

- Apply basic mathematical skills needed to understand concepts in electromagnetism
- Demonstrate the understanding on the fundamental concepts of electromagnetism
- Use Maxwell equations to explain EM wave propagation

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho_v}{\epsilon} & (\text{Gauss' Law}) \\ \nabla \cdot \mathbf{H} &= 0 & (\text{Gauss' Law for Magnetism}) \\ \nabla \times \mathbf{E} &= -\mu \frac{\partial \mathbf{H}}{\partial t} & (\text{Faraday's Law}) \\ \nabla \times \mathbf{H} &= \mathbf{J} + \epsilon \frac{\partial \mathbf{E}}{\partial t} & (\text{Ampere's Law})\end{aligned}$$



- Ability of solving problems in electromagnetic applications

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Review of Vector Analysis

Physical quantities:

Scalar quantities: Quantities having magnitude only.

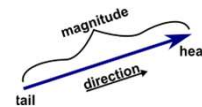
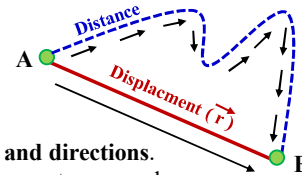
- Mass, distance, time, temperature, energy.

Vector quantities: Quantities having both magnitude and directions.

- Weight, displacement, velocity, acceleration, linear momentum, angular momentum, electric field, magnetic field, dipoles.

Geometrical representation of a vector:

Vectors can be represented graphically as arrows

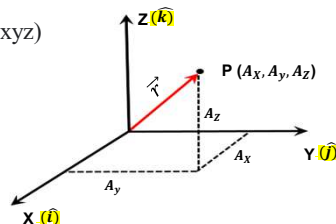


Algebraic representation of a vector:

Let P be a point in the Cartesian coordinate system (xyz)

$$\vec{r} = \mathbf{r} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

↑ Magnitude
 ↑ Directions (unit vector)



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Rules Associated with Vector Algebra

Suppose A and B are two arbitrary vectors.

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}$$

$$\mathbf{B} = B_x \hat{\mathbf{i}} + B_y \hat{\mathbf{j}} + B_z \hat{\mathbf{k}}$$

Vector addition and subtraction: $\mathbf{A} \pm \mathbf{B} = (A_x \pm B_x)\hat{\mathbf{i}} + (A_y \pm B_y)\hat{\mathbf{j}} + (A_z \pm B_z)\hat{\mathbf{k}}$

Scalar multiplication or dot product: $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta$

Here θ is the angle between two vectors. The dot product is commutative ($\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$)

Vector multiplication or cross product: $\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}|\sin\theta \hat{\mathbf{n}}$

Where $\hat{\mathbf{n}}$ is a unit vector pointing perpendicular to the plane A and B. The dot product is not commutative ($\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$)

Cross product in determinant form:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \rightarrow \mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y)\hat{\mathbf{i}} - (A_x B_z - A_z B_x)\hat{\mathbf{j}} + (A_x B_y - A_y B_x)\hat{\mathbf{k}}$$

Example: $\mathbf{A} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$, $\mathbf{B} = 1\hat{\mathbf{i}} + 0\hat{\mathbf{j}} - 1\hat{\mathbf{k}}$ 5

Answer: $\mathbf{A} \times \mathbf{B} = -3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

Cross product is based on the nature of our 3-dimensional space.

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Position, unit and separation vectors

Position vector:

The location of a point in three dimensions can be described by Cartesian coordinates (xyz). The vector to that point from the origin is called the position vector $\vec{r}$

$$\vec{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}} \quad \leftarrow \text{position vector}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2} \quad \leftarrow \text{magnitude of the position vector}$$

Unit vector:

Unit vector is obtained by dividing a vector by its own magnitude

$$\text{EX-} \quad \hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}}{\sqrt{x^2 + y^2 + z^2}}$$

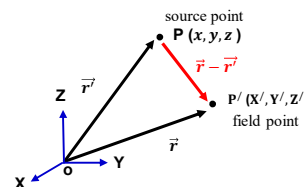
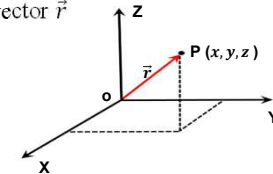
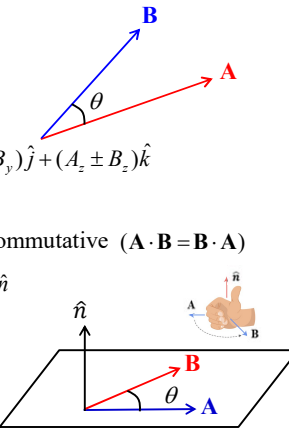
Separation vector:

In electrodynamics one frequently encountered problems involving two points

Example- source point ($\mathbf{r}'$) and field point ($\mathbf{r}$)

Separation vector from source point to field point = $\vec{r} - \vec{r}'$

$$\text{Unit vector along the Separation vector} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}$$



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Differential calculus (Gradient, Divergence and Curl)

Gradient (of a scalar field):

Suppose we have a scalar function of three variables, say $\varphi(x, y, z)$. Then the gradient of the scalar function is denoted by $\nabla \varphi$; where ∇ denotes del operator (or nabla)

In a 3D **Cartesian coordinate system** it is denoted by, $\left(\text{Del operator ; } \nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right)$

$$\nabla \varphi = \text{grad } \varphi = \frac{\partial \varphi}{\partial x} \hat{i} + \frac{\partial \varphi}{\partial y} \hat{j} + \frac{\partial \varphi}{\partial z} \hat{k} = \frac{\partial \varphi}{\partial x_i} \hat{e}_i \leftarrow \text{The gradient of a scalar field is a vector}$$



The divergence (of a vector field):

The divergence of vector F can be expressed as the dot product of ∇ and $\mathbf{F}$

Divergence of $\mathbf{F}$ ($\text{div } \mathbf{F}$) is a measure of how much the vector spread out from the origin.

$$\nabla \cdot \mathbf{F} < 0 \quad \nabla \cdot \mathbf{F} > 0 \quad \nabla \cdot \mathbf{F} = 0$$

In a 3D **Cartesian coordinate system**,

$$\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \leftarrow \text{The divergence computes a scalar quantity}$$

where, $\mathbf{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$



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Differential calculus (Gradient, Divergence and Curl)

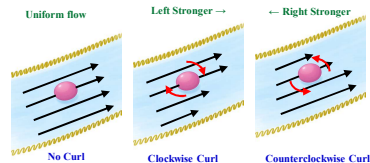
The curl (of a vector field):

The curl of a vector field $\mathbf{F}$ can be expressed as the cross product of ∇ and $\mathbf{F}$
Curl of vector $\mathbf{F}$ is measure of how much the vector $\mathbf{F}$ curls around a point.

Determinant form of the curl

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \leftarrow \text{The curl of a vector field } F \text{ is a vector}$$

$$\left(\text{Del operator ; } \nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \text{ Here, } \mathbf{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$



Exercise

1. Determine the divergence of the vector field : $F = (x^2y, yz^2, zx^2)$

$$\text{ANS} = (2xy + z^2 + x^2)$$

2. Compute the curl of the vector field : $G = (yz, xz, xy)$

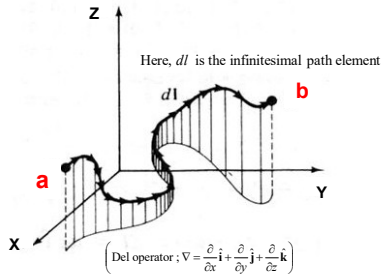
$$\text{ANS} = (0, 0, 0)$$

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Fundamental Theorem for the Gradient:

The gradient theorem says that a line integral through a gradient field can be evaluated by evaluating the original scalar field at the endpoints of the curve.



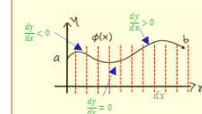
Note:

1. The line integral is path-independent
2. For a closed path, the integral evaluates to zero

$$\oint \nabla \phi \, dl = 0$$

Close path integration

For a two-dimensional system



$$\int_a^b \frac{d\phi}{dx} dx = \phi(b) - \phi(a)$$

$$\int_a^b \nabla \phi \, dl = \phi(b) - \phi(a)$$

or

$$\int_a^b (\text{grad } \phi) \, dl = \phi(b) - \phi(a)$$

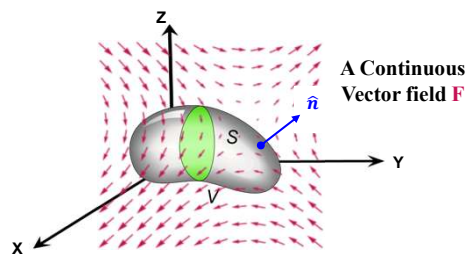
Also known as the
fundamental theorem
of calculus for line
integral

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Fundamental Theorem for Divergence: Gauss's Theorem

The fundamental theorem for the divergence states that, the volume integral of divergence of a vector field $\vec{F}$ (or $\mathbf{F}$) over a volume is equal to surface integral of $\mathbf{F}$ over a closed surface enclosing the volume.



Note:

1. The surface can have an arbitrary shape
2. The surface must be a closed surface, enclosing a finite region of space.
3. The vector field must be continuous

$$\int_V (\nabla \cdot \mathbf{F}) \, dv = \oint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds$$

or

$$\int_V (\text{div } \mathbf{F}) \, dv = \oint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds$$

Also known as the
Gauss's divergence
theorem

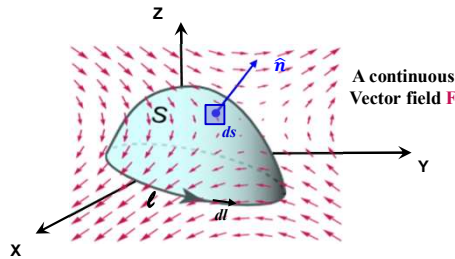
Where, $\hat{\mathbf{n}}$ is a unit vector normal to the surface

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Fundamental Theorem for the Curl: Stoke's Theorem

The fundamental theorem for curls states that; the surface integral of curl of a vector field $\vec{F}$ (or $\mathbf{F}$) over a surface is equal to line integral of $\mathbf{F}$ over the boundary of surface.



Note:

1. The boundary of the surface must be a closed loop
2. The unit normal vector must align with the right-hand rule for the boundary curve.
3. The vector field must be continuous

$$\int_S (\nabla \times \mathbf{F}) \cdot \hat{n} \, ds = \oint_L \mathbf{F} \cdot d\mathbf{l}$$

or

$$\int_S (\text{curl } \mathbf{F}) \cdot \hat{n} \, ds = \oint_L \mathbf{F} \cdot d\mathbf{l}$$

← *Stoke's theorem*

Where, $\hat{n}$ is a unit vector normal to the surface

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Second derivatives (successive applications)

$$1) \quad \nabla \cdot \nabla \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi, \quad \nabla^2 = \text{Laplacian operator}$$

$$2) \quad \nabla \times \nabla \phi = 0 \quad \leftarrow \quad \text{Curl of a gradient is always zero}$$

$$3) \quad \nabla \cdot (\nabla \times \mathbf{V}) = 0 \quad \leftarrow \quad \text{Divergence of a curl is always zero}$$

$$4) \quad \nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - (\nabla \cdot \nabla) \mathbf{V} = \nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V}$$

Here, ϕ is a scalar and $\mathbf{V}, \nabla \phi, (\nabla \times \mathbf{V})$ are vectors

Integral calculus (Summary)

Line integral: $\int_A^B \mathbf{F} \cdot d\mathbf{l} \rightarrow$ The circulation of $\mathbf{F}$ around any loop $\rightarrow \oint \mathbf{F} \cdot d\mathbf{l} = 0$.

($\mathbf{F}$ is a vector function: $d\mathbf{l}$ is infinitesimal displacement vector)

Surface integral: $\int \mathbf{F} \cdot \hat{n} \, ds$ (mostly used surface integral, *flux*)

($\mathbf{F}$ is a vector function: $\hat{n}$ is a unit vector perpendicular to the area: ds is infinitesimal surface element)

Volume integral: $\int \phi \, dv$

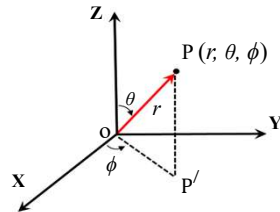
(ϕ is a scalar function: dv is infinitesimal volume element)

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Spherical Polar Coordinates

The spherical polar coordinates of a point P are defined by $(r; \theta, \phi)$ are called radial distance, polar angle and azimuthal angle respectively



Spherical and Cartesian coordinates:

The relation between two coordinates systems

$$x = r \sin \theta \cdot \cos \phi$$

$$y = r \sin \theta \cdot \sin \phi$$

$$z = r \cos \theta$$

Range of variables in spherical coordinates:

$$r \longrightarrow (0 - r); \quad \theta \longrightarrow (0 - \pi); \quad \phi \longrightarrow (0 - 2\pi)$$

Three unit vectors $\hat{r}, \hat{\theta}, \hat{\phi}$ of the spherical polar coordinates can be written in terms of Cartesian coordinates $(\hat{x}, \hat{y}, \hat{z})$

$$\hat{r} = \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}$$

$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

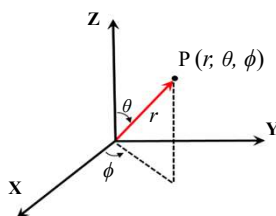
$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

NOTE — $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\hat{r} = \frac{\frac{\partial \vec{r}}{\partial r}}{\left| \frac{\partial \vec{r}}{\partial r} \right|} \quad \hat{\theta} = \frac{\frac{\partial \vec{r}}{\partial \theta}}{\left| \frac{\partial \vec{r}}{\partial \theta} \right|} \quad \hat{\phi} = \frac{\frac{\partial \vec{r}}{\partial \phi}}{\left| \frac{\partial \vec{r}}{\partial \phi} \right|}$$

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Spherical Polar Coordinates, Cont.



The infinitesimal displacement $d\mathbf{l}$:

$$d\mathbf{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

The infinitesimal surface element ds :

$$ds = r^2 \sin \theta d\theta d\phi \hat{r}; \text{ or } ds = r dr d\theta \hat{\theta}$$

(when r is constant) (when θ is constant)

The infinitesimal volume element dv :

$$dv = r^2 \sin \theta dr d\theta d\phi$$

Gradient: $\nabla \Phi = \frac{\partial \Phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial \Phi}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \phi} \hat{\phi}$

Divergence: $\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$

Curl: $\nabla \times \mathbf{F} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta F_\phi) - \frac{\partial F_\theta}{\partial \phi} \right] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial}{\partial r} (r F_\phi) \right] \hat{\theta}$
 $+ \frac{1}{r} \left[\frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right] \hat{\phi}$

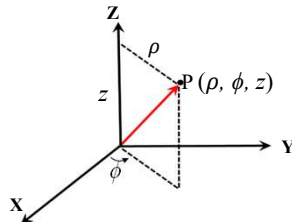
Laplacian: $\nabla^2 \Phi = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial \Phi}{\partial r}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial \Phi}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Phi}{\partial \phi^2}$

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Cylindrical Coordinates

The cylindrical coordinates of a point P are defined by (ρ, ϕ, z) as follows. Here ρ is the distance to point P from the Z axis, ϕ has the same meaning as in spherical coordinates and Z is same as in Cartesian coordinates



Cylindrical and Cartesian coordinates:

The relation between two coordinates systems

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

Range of variables in spherical coordinates:

$$\rho \rightarrow (0 - \infty); \quad \phi \rightarrow (0 - 2\pi); \quad z \rightarrow (-\infty - \infty)$$

Three unit vectors $\hat{\rho}, \hat{\phi}, \hat{z}$ of the spherical polar coordinates can be written in terms of Cartesian coordinates $(\hat{x}, \hat{y}, \hat{z})$

$$\hat{\rho} = \cos \phi \hat{x} + \sin \phi \hat{y}$$

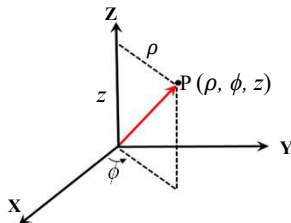
$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

$$\hat{z} = \hat{z}$$

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Cylindrical Polar Coordinates, Cont.



The infinitesimal displacement dl:

$$d\mathbf{l} = d\rho \hat{\rho} + \rho d\phi \hat{\phi} + dz \hat{z}$$

The infinitesimal surface element ds:

$$ds = \rho d\phi dz \hat{\rho}; \text{ or } ds = \rho d\rho d\phi \hat{z}$$

(when ρ is constant) (when z is constant)

The infinitesimal volume element dv:

$$dv = \rho d\rho d\phi dz$$

Gradient:
$$\nabla \Phi = \frac{\partial \Phi}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial \Phi}{\partial \phi} \hat{\phi} + \frac{\partial \Phi}{\partial z} \hat{z}$$

Divergence:
$$\nabla \cdot \mathbf{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_{\rho}) + \frac{1}{\rho} \frac{\partial F_{\phi}}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

Curl:
$$\nabla \times \mathbf{F} = \left[\frac{1}{\rho} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_{\phi}}{\partial z} \right] \hat{\rho} + \frac{1}{\rho} \left[\frac{\partial F_{\rho}}{\partial z} - \frac{\partial F_z}{\partial \rho} \right] \hat{\phi} + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho F_{\phi}) - \frac{\partial F_{\rho}}{\partial \phi} \right] \hat{z}$$

Laplacian:
$$\nabla^2 \Phi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 \Phi}{\partial \phi^2} \right) + \frac{\partial^2 \Phi}{\partial z^2}$$

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